

ReconUNet: Learning to Reconstruct Clean Covariance Matrix under Low SNR, Multipath, and Array Imperfections

Itamar Almog , and Anthony J. Weiss , *Life Fellow, IEEE*

Abstract—Direction-of-Arrival (DoA) estimation is a fundamental task in array signal processing, it is known to degrade in practical regimes characterized by low signal-to-noise ratios, limited snapshots, source coherence, and array mismatches such as gain/phase errors, sensor position bias, and mutual coupling. We propose a data-driven front-end that restores a subspace-faithful covariance prior to applying classical estimators. The method ingests multi-lag autocorrelation features and uses a U-Net backbone with lightweight convolutional and Multilayer-Perceptron (MLP) heads to reconstruct a denoised, mismatch-compensated covariance matrix; its Eigen-Decomposition (EVD) is produced explicitly and supplied to downstream algorithms. Training combines a mean-squared reconstruction loss with a subspace-similarity objective that aligns the predicted and reference signal subspaces. Comprehensive experiments cover varying numbers of sources, a broad span of Signal to Noise Ratio (SNR), both coherent and non-coherent conditions, and realistic levels of array imperfections. Using root-mean-square angular error as the primary metric, we benchmark beamforming, Minimum Variance Distortionless Response (MVDR), multiple signal classification (MUSIC), Root-MUSIC, Estimation of signal parameters via rotational invariant techniques (ESPRIT) and Unitary-ESPRIT with and without the proposed front-end. A single trained model generalizes across operating conditions and consistently improves downstream accuracy relative to applying the classical methods directly to raw sample statistics. These results indicate that learning to reconstruct the covariance—rather than regressing angles end-to-end—provides a practical, plug-and-play enhancement to classical DoA pipelines. By restoring separation between signal and noise subspaces under coherence and model mismatch, the approach yields robust, accurate estimates while preserving compatibility with standard subspace-based techniques and producing an explicit EVD useful for subsequent processing.

Index Terms—Direction-of-Arrival (DoA), array signal processing, deep learning, subspace methods

I. INTRODUCTION

Direction-of-arrival (DoA) estimation is central to array signal processing across wireless communications, radar, sonar, acoustics, and remote sensing [1], [2]. Arrays provide spatial filtering—separating signals overlapping in frequency but arriving from different directions—enabling reliable links, interference suppression, and localization [3].

Classical DoA estimators fall broadly into two families. **Beamformers** scan power over angle, from conventional (Bartlett) to adaptive designs such as Capon’s Minimum Variance Distortionless Response (MVDR), which minimizes output power subject to a distortionless constraint [4], [5]. **Subspace methods** recast DoA as a geometry problem on the array manifold: Multiple Signal Classification (MUSIC)

searches for steering vectors orthogonal to the estimated noise subspace [6]; Estimation of signal parameters via rotational invariant techniques (ESPRIT) uses translational invariance to recover inter-subarray phase shifts without a spectral search [7]; Unitary ESPRIT incorporates forward-backward processing in a real-valued pipeline [8], [9]; Root-MUSIC replaces spectral scans with polynomial rooting for ULAs [10].

Despite their maturity, classical approaches degrade with few snapshots (noisy covariance), low SNR (weak subspace separation), coherent or highly correlated/multipath (rank deficiency), and array/model mismatch (gain/phase errors, element bias, mutual coupling). Remedies such as spatial smoothing, forward-backward averaging, and self-calibration help but are scenario-specific [11]–[14]. The maturity of array processing theory affords rigorous lower bounds for benchmarking estimators. The most known among these is the Cramér–Rao lower bound (CRLB), which specifies the minimum achievable variance of any unbiased DoA estimator under the assumed data model and SNR, and thus serves as a canonical reference for asymptotic (high-SNR) performance [3], [15]. Complementing the CRLB, the Ziv–Zakai bound (ZZB) characterizes the transition to the threshold region by accounting for large-error events and closely spaced sources [16].

Over the past decade, deep learning has matured its able to provide end-to-end systems that learn hierarchical representations and optimization objectives directly from data [17], [18]. In signal processing, model-based deep learning has bridged classical inductive biases with data-driven training, yielding architectures that encode domain structure while adapting to real-world deviations [19]. Within array processing and beamforming, recent surveys chart substantial gains across detection, localization, and interference suppression when learning is aligned with physics and deployment constraints [20].

Recent **learning-based** DoA methods pursue three directions. (i) *Spectrum learning*: networks output a spatial spectrum followed by peak-detector; robustness has been shown under imperfections and low SNR when train-test conditions align [21]–[24]. (ii) *Learning the statistics*: covariance-reconstruction/subspace-guided models denoise the sample covariance and preserve separability (e.g., SubspaceNet; DAMUSIC) while remaining compatible with classical pipelines [25], [26]. (iii) *Direct predictors*: models regress or classify angles from covariance-like inputs (e.g., CNNs, DOA-ViT) with strong low-SNR performance, though handling variable

source counts and coherence can be challenging [27]–[29].

This work targets a persistent gap: a single, subspace-guided preprocessing model that reconstructs a clean covariance and its Eigen-Value-Decomposition (EVD) from noisy, imperfect measurements, so that Beamforming, MVDR, MUSIC, Root-MUSIC, and ESPRIT operate in their natural regime across low-SNR conditions, coherent/highly correlated and non-coherent mixtures, and array imperfections—without per-scenario re-tuning. Unlike recent deep-learning-based DoA estimators, which typically require retraining or fine-tuning for specific SNR ranges, or array imperfections, the proposed *ReconUNet* employs a single frozen set of weights that generalizes across all these variations simultaneously. Following the “learn the statistics, keep the physics” principle, the network consumes a stack of multi-lag autocorrelations that is intrinsically informative for subspace recovery and less sensitive to sample length than raw snapshots. A U-Net backbone with skip connections [30], coupled to lightweight convolutional heads, predicts a Hermitian positive-definite covariance together with a clean eigendecomposition. Training couples a covariance-fidelity objective with a subspace-similarity objective, encouraging both statistical accuracy and preservation of the signal–noise geometry on which classical estimators rely. At inference, the reconstructed covariance replaces the sample estimate, and the predicted eigen-structure can be used directly when convenient; classical pipelines then run unchanged, retaining interpretability and avoiding on-grid biases typical of spectrum-classification networks.

The contributions of this work are: (1) A preprocessing network that maps multi-lag autocorrelations to a clean covariance and eigendecomposition; (2) A dual-objective training scheme combining covariance fidelity with subspace preservation; and (3) A practical by-product in the form of a predicted eigendecomposition, enabling classical processing without online EVD.

Organization. Section II states the array/signal models and problem setup. Section III details the subspace-guided preprocessing framework and training. Section IV describes datasets, baselines, and evaluation. Section V reports accuracy, robustness, and generalization results. Section VI concludes with limitations and future directions.

II. SIGNAL MODEL

Section outline. We first introduce the baseline (ideal) model in Section II-A, then formalize specular multipath/coherent sources in Section II-B, incorporate practical array imperfections in Section II-C, and finally state the problem formulation in Section II-D. The array geometry and angle convention are illustrated in Fig. 1.

A. Baseline (Ideal) Model

We consider an array of N sensors with inter-element spacing d . At snapshot $t \in \{1, \dots, T\}$, the array measurement is $\mathbf{x}(t) \in \mathbb{C}^{N \times 1}$ and the stacked data matrix is $\mathbf{X} = [\mathbf{x}(1), \dots, \mathbf{x}(T)] \in \mathbb{C}^{N \times T}$. There are K emitters with directions of arrival (DoAs) $\boldsymbol{\theta} = [\theta_1, \dots, \theta_K]$, where θ is measured from the array axis (broadside = 90°). Let $\mathbf{s}(t) = [s_1(t), \dots, s_K(t)]^T \in \mathbb{C}^{K \times 1}$

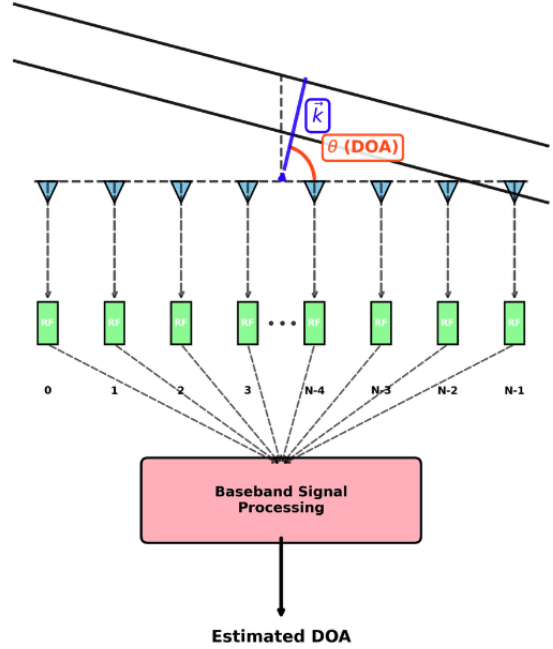


Fig. 1: Uniform linear array (ULA) geometry and angle convention. Element spacing is $d = \lambda/2$, and the DoA $\theta \in [30^\circ, 150^\circ]$ is measured from the array axis (broadside = 90°).

denote the source vector and $\mathbf{n}(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I}_N)$ spatially white noise. The steering vector (cosine convention) is

$$\mathbf{a}(\theta) = [\exp(j \mathbf{k}(\theta)^T \mathbf{r}_0), \dots, \exp(j \mathbf{k}(\theta)^T \mathbf{r}_{N-1})]^T \in \mathbb{C}^{N \times 1}. \quad (1)$$

$$\mathbf{k}(\theta) = \frac{2\pi}{\lambda} \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \quad \mathbf{r}_n \in \mathbb{R}^2. \quad (2)$$

and the steering matrix is

$$\mathbf{A}(\boldsymbol{\theta}) = [\mathbf{a}(\theta_1), \dots, \mathbf{a}(\theta_K)] \in \mathbb{C}^{N \times K}. \quad (3)$$

Element-wise,

$$[\mathbf{a}(\theta)]_n = \exp(j \mathbf{k}(\theta)^T \mathbf{r}_n), \quad n = 0, \dots, N-1. \quad (4)$$

Under linear superposition, the received signal at time t is

$$\mathbf{x}(t) = \mathbf{A}(\boldsymbol{\theta}) \mathbf{s}(t) + \mathbf{n}(t). \quad (5)$$

Define the source and array covariances

$$\mathbf{R}_s \triangleq \mathbb{E}\{\mathbf{s}(t) \mathbf{s}^H(t)\} \succeq \mathbf{0}, \quad \mathbf{R}_x \triangleq \mathbb{E}\{\mathbf{x}(t) \mathbf{x}^H(t)\}, \quad (6)$$

so that

$$\mathbf{R}_x = \mathbf{A}(\boldsymbol{\theta}) \mathbf{R}_s \mathbf{A}^H(\boldsymbol{\theta}) + \sigma_n^2 \mathbf{I}_N, \quad (7)$$

and the sample covariance matrix (SCM)

$$\hat{\mathbf{R}}_x = \frac{1}{T} \mathbf{X} \mathbf{X}^H. \quad (8)$$

If $\text{rank}(\mathbf{R}_s) = K$ (mutually non-coherent sources), $\mathbf{R}_x$ splits into a K -dimensional signal subspace and an $(N - K)$ -dimensional noise subspace.

B. Specular Multipath / Coherent Sources

a) *Specular multipath model.*: A dominant emitter can produce mirror-like replicas of the line-of-sight waveform via reflections. Each replica has excess path length $\Delta\ell_k$ expressed as group delay

$$\tau_k = \frac{\Delta\ell_k}{c}. \quad (9)$$

Let L_k be the path attenuation in dB, with linear amplitude gain

$$\alpha_k = 10^{-L_k/20}, \quad (10)$$

and let f_c denote the carrier frequency. With a phase offset ϕ_k due to the reflection coefficient, the complex-baseband per-path gain is

$$g_k = \alpha_k e^{j\phi_k} e^{-j2\pi f_c \tau_k}. \quad (11)$$

b) *Correlation constraint on delay.*: Let B denote the essential baseband bandwidth of the direct signal $s_1(t)$ and $r_1(\Delta) = \mathbb{E}\{s_1(t)s_1^*(t - \Delta)\}$ its autocorrelation. The normalized envelope

$$\gamma(\tau) \triangleq \frac{|r_1(\tau)|}{r_1(0)} \quad (12)$$

decays on the correlation-time scale $T_{\text{coh}} \approx 1/B$. To remain in the highly correlated/specular regime,

$$0 \leq \tau_k \leq \tau_{\text{max}}, \quad \tau_{\text{max}} \approx \frac{1}{B}. \quad (13)$$

c) *Received snapshot and covariance.*: Consider one direct source $s(t)$ from angle θ and one specular replica with parameters $(\tau, \theta^{\text{mp}}, g)$ as above. Using the notation from the baseline model, the array snapshot for single source is

$$\mathbf{x}(t) = \mathbf{a}(\theta) s(t) + \mathbf{a}(\theta^{\text{mp}}) g s(t - \tau) + \mathbf{n}(t). \quad (14)$$

d) *Covariance expansion.*: Let $P \triangleq \mathbb{E}\{|s(t)|^2\}$ and $r(\Delta) \triangleq \mathbb{E}\{s(t)s^*(t - \Delta)\}$. The array covariance $\mathbf{R}_x = \mathbb{E}\{\mathbf{x}(t)\mathbf{x}^H(t)\}$ expands to

$$\mathbf{R}_x = P \mathbf{a} \mathbf{a}^H + |g|^2 P \mathbf{b} \mathbf{b}^H + g P \gamma \mathbf{a} \mathbf{b}^H + g^* P \gamma^* \mathbf{b} \mathbf{a}^H + \sigma_n^2 \mathbf{I}_N. \quad (15)$$

When $\tau < 1/B$ we typically have $|r(\tau)| \leq P$, so the replica is highly correlated with the direct path.

e) *Rank-1 collapse and the effective steering vector.*: Well inside the correlation window, write $r(\tau) = P\gamma$ with $|\gamma| \approx 1$ and $\gamma \approx e^{j\varphi}$. Denote $a(\theta) \triangleq \mathbf{a}$ and $a(\theta^{\text{mp}}) \triangleq \mathbf{b}$. The signal component of the covariance (excluding $\sigma_n^2 \mathbf{I}_N$) is

$$R_{x,\text{sig}} = P \mathbf{a} \mathbf{a}^H + |g|^2 P \mathbf{b} \mathbf{b}^H + g P \gamma \mathbf{a} \mathbf{b}^H + g^* P \gamma^* \mathbf{b} \mathbf{a}^H. \quad (16)$$

This factorizes as

$$R_{x,\text{sig}} = P (a + g \gamma b) (a + g \gamma b)^H \triangleq P v_{\text{eff}} v_{\text{eff}}^H, \quad (17)$$

with the effective steering vector

$$v_{\text{eff}} = a(\theta) + g e^{j\varphi} a(\theta^{\text{mp}}). \quad (18)$$

Thus, for coherent specular multipath ($|\gamma| = 1$), the signal covariance is exactly rank 1 (outer product). For $|\gamma| \lesssim 1$, it is nearly rank 1.

C. Array Imperfections

a) *Geometry and nominal manifold.*: Nominal sensor locations are

$$\mathbf{r}_n^{(0)} = \begin{bmatrix} nd \\ 0 \end{bmatrix}, \quad n = 0, \dots, N-1, \quad (19)$$

with $k = \frac{2\pi}{\lambda}$ and $\mathbf{k}(\theta) = k[\cos \theta, \sin \theta]^T$. The nominal steering entry is

$$[\mathbf{a}_0(\theta)]_n = \exp(j \mathbf{k}(\theta)^T \mathbf{r}_n^{(0)}) = \exp(j \pi n \cos \theta). \quad (20)$$

b) *Position errors (element misplacement).*: Each element is perturbed by

$$\boldsymbol{\varepsilon}_n \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{pos}}^2 \mathbf{I}_2), \quad (21)$$

giving the actual position and perturbed manifold

$$\mathbf{r}_n = \mathbf{r}_n^{(0)} + \boldsymbol{\varepsilon}_n, \quad [\mathbf{a}_{\text{pos}}(\theta)]_n = \exp(j \mathbf{k}(\theta)^T \mathbf{r}_n). \quad (22)$$

This direction-dependent phase distortion induces steering-vector mismatch, DoA bias, and mainlobe broadening.

c) *Per-element gain/phase errors.*: Hardware variability which is representing a moderate quality mechanical design, fabrication and assembly procedures is modeled by a diagonal matrix

$$\mathbf{G} = \text{diag}(g_n e^{j(\pi/180) \phi_n})_{n=0}^{N-1}, \quad g_n \sim \mathcal{U}[g_{\min}, g_{\max}], \quad \phi_n \sim \mathcal{U}[\phi_{\min}, \phi_{\max}] \text{ (degrees)}. \quad (23)$$

This introduces amplitude/phase imbalance and loss of nominal symmetry.

d) *Mutual coupling (Toeplitz, reciprocal).*: Electromagnetic coupling due to moderate quality production procedure is captured by a Toeplitz matrix

$$\mathbf{M} = \mathbf{I}_N + \mathbf{E}, \quad \mathbf{E}[i, j] = \begin{cases} 0, & i = j, \\ c_{|i-j|}, & i \neq j, \end{cases} \quad (24)$$

with geometrically decaying off-diagonals

$$\begin{aligned} c_k &= \gamma^k v_k, \quad k = 1, \dots, N-1, \\ \gamma &= \rho e^{j\varphi_c}, \\ v_k &\sim \mathcal{U}[1 - \frac{\delta}{2}, 1 + \frac{\delta}{2}]. \end{aligned} \quad (25)$$

This reciprocal, distance-ordered model captures non-diagonal sensor interaction and loss of centro-symmetry relevant to Unitary ESPRIT.

e) *Composite operator, distorted manifold, and covariance.*: Collecting hardware effects,

$$\begin{aligned} \mathbf{H} &= \mathbf{M} \mathbf{G}, \\ \tilde{\mathbf{a}}(\theta) &= \mathbf{H} \mathbf{a}_{\text{pos}}(\theta), \\ \tilde{\mathbf{A}}(\boldsymbol{\theta}) &= [\tilde{\mathbf{a}}(\theta_1), \dots, \tilde{\mathbf{a}}(\theta_K)]. \end{aligned} \quad (26)$$

Component-wise,

$$[\tilde{\mathbf{a}}(\theta)]_i = \sum_{n=1}^N (\mathbf{M})_{i,n} g_n \exp(j(\pi/180) \phi_n) \exp(j \mathbf{k}(\theta)^T \mathbf{r}_n), \quad (27)$$

with $i = 0, \dots, N-1$. The observation and covariance become

$$\mathbf{x}(t) = \tilde{\mathbf{A}}(\boldsymbol{\theta}) \mathbf{s}(t) + \mathbf{n}(t), \quad \mathbf{R}_x = \tilde{\mathbf{A}} \mathbf{R}_s \tilde{\mathbf{A}}^H + \sigma_n^2 \mathbf{I}_N. \quad (28)$$

f) *Implications (qualitative).*:

- Steering mismatch and bias: position errors and gain/phase imbalance tilt $\tilde{\mathbf{a}}(\theta)$ away from $\mathbf{a}_0(\theta)$, biasing DoA estimates and degrading resolution.
- Subspace leakage: coupling mixes sensor outputs; even for uncorrelated sources, columns of $\tilde{\mathbf{A}}$ are not orthogonal under the noise metric, shrinking the signal-noise eigengap and flattening MUSIC-type spectra.
- Loss of symmetry/centro-Hermiticity: Toeplitz coupling and per-element phases break structures exploited by Unitary ESPRIT and forward-backward averaging, weakening rotational invariance and increasing variance.

D. Problem Formulation

a) *Objective.*: Given T narrowband, far-field snapshots $\mathbf{X} = [\mathbf{x}(1), \dots, \mathbf{x}(T)] \in \mathbb{C}^{N \times T}$ acquired by an array, estimate the set of directions of arrival (DoAs) $\boldsymbol{\theta} = \{\theta_1, \dots, \theta_K\}$. Depending on the setting, the source count K is either known (enumeration provided) or must be estimated jointly.

b) *General model.*: Snapshots obey the array models of Sections II-A, II-B, and II-C. At time t ,

$$\mathbf{x}(t) = \tilde{\mathbf{A}}(\boldsymbol{\theta})\mathbf{s}(t) + \mathbf{x}_{\text{mp}}(t) + \mathbf{n}(t), \quad (29)$$

where $\tilde{\mathbf{A}}(\boldsymbol{\theta})$ is the imperfection-distorted steering matrix (Section II-C), $\mathbf{x}_{\text{mp}}(t)$ is the contribution of coherent specular replicas (Section II-B), and $\mathbf{n}(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I}_N)$ is spatially white noise, independent of the signals. Signals may be mutually correlated or coherent. The sample covariance (SCM) and the ensemble covariance are

$$\hat{\mathbf{R}}_x = \frac{1}{T} \mathbf{X} \mathbf{X}^H, \quad \mathbf{R}_x = \mathbb{E}\{\mathbf{x}(t) \mathbf{x}^H(t)\}. \quad (30)$$

In the absence of coherence and with a matched manifold, $\mathbf{R}_x$ decomposes into a K -dimensional signal subspace and an $(N - K)$ -dimensional noise subspace; with highly correlated multipath and imperfections, subspace leakage, rank collapse, and steering mismatch can occur.

c) *Estimation task and metric.*: Let $\mathcal{G}$ denote a chosen decision rule (e.g., beamforming scan, MUSIC/Root-MUSIC spectrum, ESPRIT). The DoA estimator produces a set of angles $\hat{\boldsymbol{\theta}} = \mathcal{G}(\mathbf{X})$ (or $\mathcal{G}(\hat{\mathbf{R}}_x)$). Performance is evaluated with a permutation-invariant set error; writing Π_K for the set of $K \times K$ permutation matrices,

$$\text{RMSE}^2(\hat{\boldsymbol{\theta}}, \boldsymbol{\theta}) = \frac{1}{K} \min_{\mathbf{P} \in \Pi_K} \|\hat{\boldsymbol{\theta}} - \mathbf{P}\boldsymbol{\theta}\|_2^2, \quad (31)$$

and we report RMSE in degrees over test scenes.

III. PROPOSED METHOD

A. Overview

We propose a subspace-guided preprocessing network (RecoNet) that maps raw snapshots to two physics-consistent products: (i) a denoised, Hermitian covariance (“clean SCM”), and (ii) a self-consistent eigendecomposition (“clean EVD”). The network *does not* replace classical DoA estimators; it repairs second-order structure so that MUSIC/Root-MUSIC/ESPRIT operate closer to their ideal regime under low SNR, coherent

multipath, and array imperfections. See Fig. 2 for the training flow and Fig. 3 for the inference pipeline.

B. Feature Representation: Multi-Lag Autocorrelation

Given snapshots $\{\mathbf{x}(t)\}_{t=0}^{T-1}$, form short-range temporal autocorrelations up to lag τ :

$$\begin{aligned} \hat{\mathbf{R}}_x[\ell] &= \frac{1}{T-\ell} \sum_{t=\ell+1}^T \mathbf{x}(t) \mathbf{x}^H(t-\ell), \\ \hat{\mathbf{R}}_x[\ell] &\in \mathbb{C}^{N \times N}, \quad \ell = 0, \dots, \tau-1. \end{aligned} \quad (32)$$

Stack the τ complex matrices along a channel axis and convert to a real tensor via

$$\begin{aligned} \text{Stack}_{\Re\Im} : \mathbb{C}^{N \times N} &\rightarrow \mathbb{R}^{2N \times N}, \\ \mathbf{Z} &\mapsto \begin{bmatrix} \Re\{\mathbf{Z}\} \\ \Im\{\mathbf{Z}\} \end{bmatrix}, \end{aligned} \quad (33)$$

so the network input has shape $B \times \tau \times 2N \times N$.

Why this feature works:

- *Lower variance, same physics:* second-order statistics concentrate what subspace methods use (SCM/EVD) and average temporal noise.
- *Multipath cues:* correlated replicas that collapse rank at $\ell=0$ leave structure at small $\ell>0$, exposed by stacking lags.
- *Useful invariances:* autocorrelation removes absolute phase/time shifts; unit-norm steering yields stable dynamic range.
- *Short lags suffice:* small τ captures informative temporal structure without inflating input size.

C. Network Architecture

U-Net backbone: A compact 2D U-Net processes the lag-sensor grid with three encoder/decoder levels. Each encoder stage applies two 3×3 convolutions (with normalization and nonlinearity) and 2×2 downsampling; the decoder mirrors this with transposed convolutions and skip connections that preserve fine diagonal/Hermitian patterns. This matches our goal of repairing second-order statistics rather than regressing angles.

Anti-rectifier activation: Besides ReLU, an *anti-rectifier* concatenates positive and negative responses to preserve sign information critical for real/imag structure:

$$\text{AR}(\mathbf{X}) = [\text{ReLU}(\mathbf{X}) \parallel \text{ReLU}(-\mathbf{X})]. \quad (34)$$

The resulting backbone is illustrated in Fig. 4.

D. SCM Head: Hermitian Projection and Diagonal Loading

The decoder predicts an $N \times N$ complex matrix via the inverse of (33). We enforce Hermitianity and stabilize conditioning:

$$\hat{\mathbf{R}}_{\text{Herm}} = \frac{1}{2} (\hat{\mathbf{R}}_{\text{raw}} + \hat{\mathbf{R}}_{\text{raw}}^H), \quad \hat{\mathbf{R}} = \hat{\mathbf{R}}_{\text{Herm}} + \varepsilon \mathbf{I}_N, \quad (35)$$

with a small load $\varepsilon > 0$.

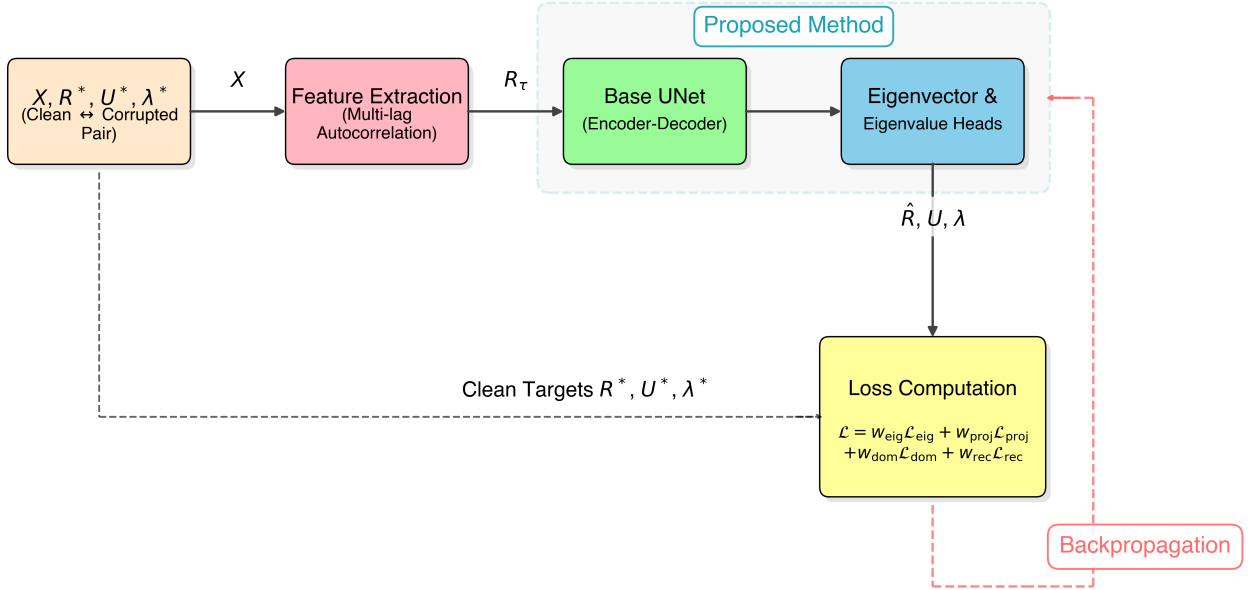


Fig. 2: Training pipeline: corrupted snapshots $\rightarrow$ lag-stack features $\rightarrow$ U-Net heads $\rightarrow$ loss computation.

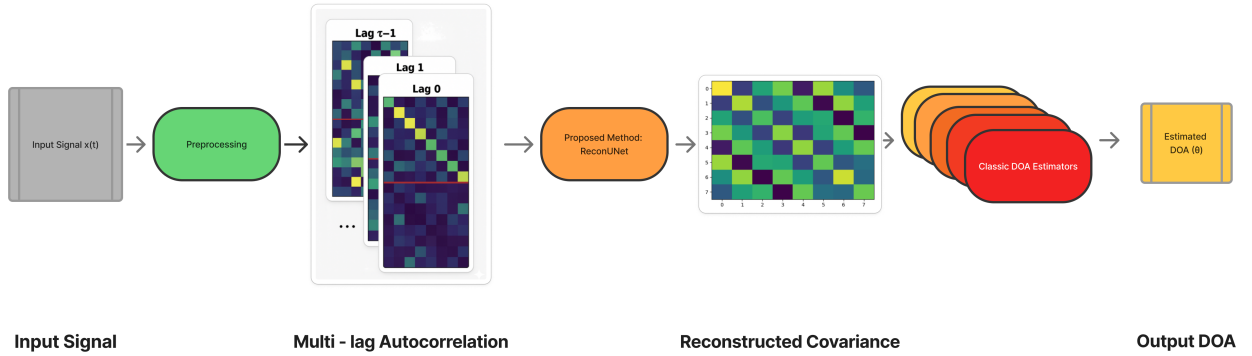


Fig. 3: End-to-end inference pipeline (multi-lag autocorrelation $\rightarrow$ EVD U-Net $\rightarrow$ classical DoA estimators).

E. EVD Heads: Spectrum and Basis with Complex QR

Eigenvalue head: Global average pooling over the $(2N \times N)$ plane followed by an MLP predicts nonnegative eigenvalues $\hat{\lambda} \in \mathbb{R}_+^N$, sorted in descending order:

$$\hat{\lambda}_\downarrow = \Pi \hat{\lambda}, \quad \hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_N. \quad (36)$$

Eigenvector head with complex QR and phase anchoring: A shallow CNN predicts $\hat{\mathbf{U}}_{\text{raw}} \in \mathbb{C}^{N \times N}$ (real/imag stack $\rightarrow$

complex). Project to the Stiefel manifold via a differentiable complex QR:

$$[\hat{\mathbf{Q}}, \hat{\mathbf{R}}] = \text{cQR}(\hat{\mathbf{U}}_{\text{raw}}), \quad \hat{\mathbf{U}} = \hat{\mathbf{Q}}, \quad \hat{\mathbf{U}}^H \hat{\mathbf{U}} = \mathbf{I}_N. \quad (37)$$

Resolve per-column phase by anchoring a reference entry to real-positive:

$$\varphi_i = \arg([\hat{\mathbf{U}}]_{1,i}), \quad \bar{\mathbf{U}} = \hat{\mathbf{U}} \text{diag}(e^{-j\varphi_1}, \dots, e^{-j\varphi_N}). \quad (38)$$

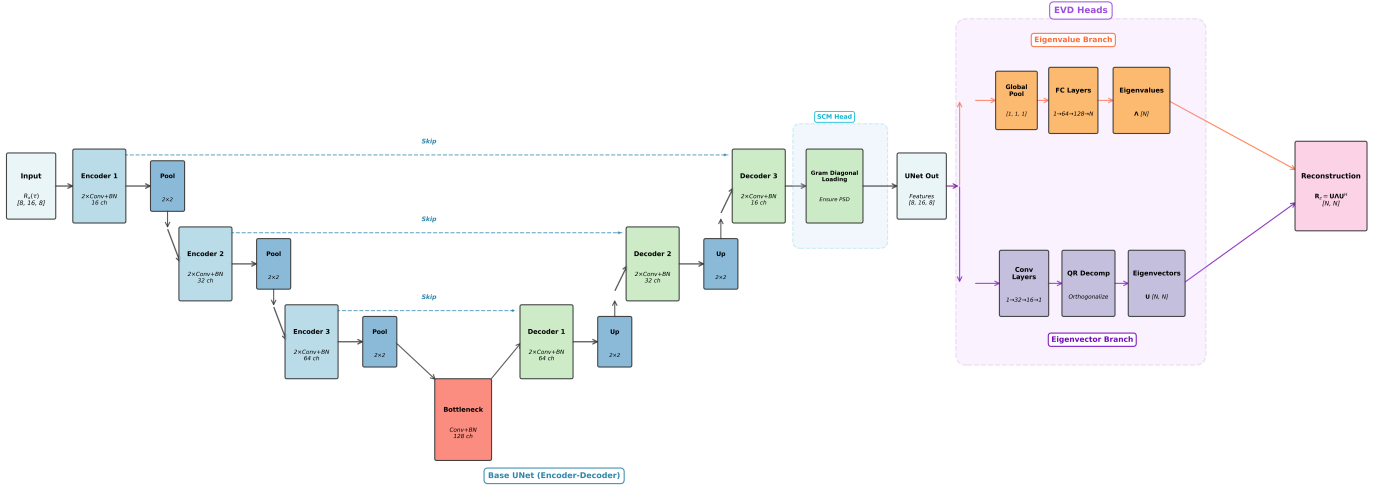


Fig. 4: U-Net backbone with skip connections, SCM head, and EVD heads.

The EVD-reconstructed covariance is

$$\hat{\mathbf{R}}_{\text{EVD}} = \bar{\mathbf{U}} \text{diag}(\hat{\lambda}_{\downarrow}) \bar{\mathbf{U}}^H. \quad (39)$$

F. Training Data Generation and Objectives

Inputs (corrupted).: For each scene we synthesize snapshots and features under the imperfect array of Section II-C, with per-scene randomization of (i) per-element gain/phase errors, (ii) small position jitter, and (iii) reciprocal Toeplitz mutual coupling. Optional correlated replicas (specular multipath) follow Section II-B. Noise power is set to the target SNR. These corrupted snapshots are converted into the lag stack (cf. Section III-B).

Targets (clean, nominal manifold).: Supervision uses a clean, imperfection-free covariance and its eigendecomposition on the nominal array manifold of Section II-A. With direct-path DoAs θ , nominal steering $\mathbf{A}_0(\theta)$, and source covariance $\mathbf{R}_s^* \succeq 0$,

$$\mathbf{R}^* = \mathbf{A}_0(\theta) \mathbf{R}_s^* \mathbf{A}_0^H(\theta), \quad (40)$$

and $\mathbf{R}^* = \mathbf{U}^* \text{diag}(\lambda^*)(\mathbf{U}^*)^H$.

Model outputs.: Given the corrupted inputs, the network predicts $\hat{\mathbf{U}} \in \mathbb{C}^{N \times N}$ (unitary via cQR), $\hat{\lambda} \in \mathbb{R}_+^N$ (sorted), and the reconstructed covariance

$$\hat{\mathbf{R}} = \hat{\mathbf{U}} \text{diag}(\hat{\lambda}) \hat{\mathbf{U}}^H. \quad (41)$$

Training objective.: A composite loss ties predictions to clean targets:

$$\mathcal{L} = w_{\text{eig}} \mathcal{L}_{\text{eig}} + w_{\text{proj}} \mathcal{L}_{\text{proj}} + w_{\text{dom}} \mathcal{L}_{\text{dom}} + w_{\text{rec}} \mathcal{L}_{\text{rec}}. \quad (42)$$

Signal-subspace alignment (projection–Frobenius) for leading K columns:

$$\mathcal{L}_{\text{proj}} = \left\| \hat{\mathbf{U}}_K \hat{\mathbf{U}}_K^H - \mathbf{U}_K^* (\mathbf{U}_K^*)^H \right\|_F^2. \quad (43)$$

Dominant-eigenvector tie (sign-robust):

$$\mathcal{L}_{\text{dom}} = \min \left(\|\hat{\mathbf{u}}_1 - \mathbf{u}_1^*\|_2^2, \|\hat{\mathbf{u}}_1 + \mathbf{u}_1^*\|_2^2 \right). \quad (44)$$

Algorithm 1: Training using subspace-guided statistics repair (EVD-only head)

Init: Lags τ ; weights $(w_{\text{eig}}, w_{\text{proj}}, w_{\text{dom}}, w_{\text{rec}})$; learning rate μ ; epochs e_{max} ; batches per epoch B .

Input: Training set $\mathcal{D} = \{(\mathbf{X}, \mathbf{R}^*, \mathbf{U}^*, \lambda^*, K)\}$, with $\mathbf{X} \in \mathbb{R}^{\tau \times 2N \times N}$ from (32)–(33).

```

1 Initialize  $\Theta$ .
2 for  $epoch \leftarrow 1$  to  $e_{\text{max}}$  do
3   Partition  $\mathcal{D}$  into mini-batches  $\{\mathcal{D}_b\}_{b=1}^B$ .
4   for  $b \leftarrow 1$  to  $B$  do
5     Load batch  $(\mathbf{X}, \mathbf{R}^*, \mathbf{U}^*, \lambda^*, K)$ .
6      $(\hat{\mathbf{U}}_{\text{raw}}, \hat{\lambda}) \leftarrow f_{\Theta}(\mathbf{X})$ .
7      $\hat{\mathbf{U}} \leftarrow \text{cQR}(\hat{\mathbf{U}}_{\text{raw}})$ ;
8     sort  $\hat{\lambda}$ .
9     Phase-anchor  $\hat{\mathbf{U}} \Rightarrow \bar{\mathbf{U}}$ .
10     $\hat{\mathbf{R}} = \bar{\mathbf{U}} \text{diag}(\hat{\lambda}) \bar{\mathbf{U}}^H$ .
11    Compute  $\mathcal{L}_{\text{proj}}, \mathcal{L}_{\text{dom}}, \mathcal{L}_{\text{eig}}, \mathcal{L}_{\text{rec}}$ .
12     $\mathcal{L} = w_{\text{proj}} \mathcal{L}_{\text{proj}} + w_{\text{dom}} \mathcal{L}_{\text{dom}} + w_{\text{eig}} \mathcal{L}_{\text{eig}} + w_{\text{rec}} \mathcal{L}_{\text{rec}}$ .
13  Update  $\Theta \leftarrow \Theta - \mu \nabla_{\Theta} \mathcal{L}$ .
```

Eigenvalue fidelity:

$$\mathcal{L}_{\text{eig}} = \frac{1}{N} \|\hat{\lambda} - \lambda^*\|_2^2. \quad (45)$$

EVD consistency (reconstruction tie):

$$\mathcal{L}_{\text{rec}} = \frac{1}{N^2} \|\hat{\mathbf{R}} - \mathbf{R}^*\|_F^2. \quad (46)$$

G. Inference and Classical Integration

At test time, compute (32), pass the stack through the network, and use either route:

- *Covariance route*: feed $\hat{\mathbf{R}}_{\text{EVD}}$ from (39) to MUSIC/Root-MUSIC/ESPRIT (with K provided or estimated externally).
- *Subspace route*: use columns of $\bar{\mathbf{U}}$ as the signal/noise basis, bypassing runtime EVD.

Angle extraction remains classical; the network focuses on repairing second-order structure.

IV. TRAINING APPROACH AND EXPERIMENTAL SETUP

Scope. We state the data synthesis, input features, supervision signals, exact training hyperparameters, baselines, metrics, and stress scenarios required to reproduce the results. Architectural details and loss definitions appear earlier.

Signal model and geometry. Unless noted otherwise: ULA with $N=8$ and half-wavelength spacing $d=\lambda/2$, narrowband around 2.45 GHz, array-axis angle convention $\theta \in [0^\circ, 180^\circ]$ (broadside = 90°), and $T=512$ snapshots per scene. SNR is defined w.r.t. the *main* source power P as $\text{SNR} = 10 \log_{10}(P/\sigma_n^2)$; σ_n^2 is set from the target. We restrict sampling and scanning to $[30^\circ, 150^\circ]$ to avoid endfire ambiguities and poor conditioning at $\{0^\circ, 180^\circ\}$ where a ULA's cosine manifold is least informative.

Dataset generation (paired *clean* $\leftrightarrow$ *corrupted*). Learning is cast as *statistics repair*. For each scene we store:

- *Target (clean)*. Ideal manifold, no multipath, no imperfections; the clean covariance $\mathbf{R}^*$ and its EVD ($\mathbf{U}^*, \boldsymbol{\lambda}^*$) serve as supervision.
- *Input (corrupted)*. The same scene rendered with additive noise at the chosen SNR, optional *correlated* specular replicas, and array imperfections (per-element gain/phase offsets, small position jitter, reciprocal Toeplitz mutual coupling) per Section II-C.

Angles and separation: direct-path DoAs are drawn from $\{30^\circ:150^\circ\}$ with a minimum 10° separation between *direct* sources; replicas are unconstrained. *Counts*: one main source plus $I \in \{0, 1, 2, 3\}$ equal-power interferers (so $\text{SIR} = 0$ dB when $I>0$), giving $K = 1 + I$. *Correlated multipath*: $M \in \{0, 1, 2, 3\}$ narrowband replicas with small delays and independent departure angles. We render both perfect and imperfect arrays, with imperfection parameters randomized per scene.

Dataset size and splits. The corpus comprises $\approx 2.0 \times 10^6$ paired examples (corrupted input $\rightarrow$ clean target), covering both perfect and imperfect arrays across the SNR schedule below. Train/validation/test partitions are fixed by seed; all reported numbers in Section V use the held-out test split.

Feature construction (network inputs). From corrupted snapshots we compute short-lag autocorrelations

$$\hat{\mathbf{R}}_x[\ell] = \frac{1}{T-\ell} \sum_{t=\ell+1}^T \mathbf{x}(t) \mathbf{x}^H(t-\ell), \quad \ell = 0, \dots, \tau-1, \quad (47)$$

then split real/imag parts to form a real tensor of shape $B \times \tau \times 2N \times N$.

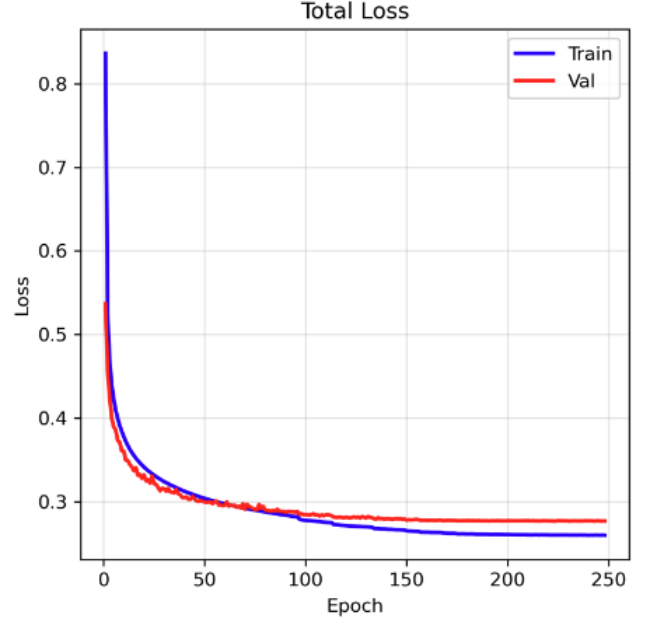


Fig. 5: Total loss over epochs on train/validation splits.

Supervision and losses. Each sample provides its ground-truth K (direct-sources count) to index the signal subspace. The training objective combines:

- projection–Frobenius *signal-subspace* alignment,
- a dominant-eigenvector consistency term (sign/phase robust),
- eigenvalue fidelity,
- and an EVD reconstruction tie to the clean covariance.

Column orthonormality is enforced via a differentiable complex-QR layer (no explicit penalty).

Exact training hyperparameters. Unless stated otherwise:

- *Model*: lag depth $\tau=8$; sensors $N=8$ (input grid $2N \times N$ via real/imag split); anti-rectifier activation; dropout enabled. The SCM head applies Hermitian symmetrization and diagonal loading with $\varepsilon=1$.
- *Optimizer*: Adam, LR 1×10^{-4} , weight decay 1×10^{-5} .
- *Scheduler*: ReduceLROnPlateau (mode=min), factor 0.7, patience 5 epochs.
- *Batching*: batch size 2048; up to 300 epochs; early-stopping patience 25 epochs (validation loss).
- *Grad clipping*: global ℓ_2 norm clipped at 1.0.
- *SNR schedule*: training sweep $\text{SNR} \in [-10, 10]$ dB; evaluation sweeps $[-20, 20]$ dB.
- *Imperfections*: randomized per scene as in Section II-C.

Inference and classical integration. At test time we form the lag-stack, apply the network, and use either:

- *Reconstructed-covariance route*: feed $\hat{\mathbf{R}}_{\text{clean}}$ into Bartlett/MVDR/MUSIC.
- *Subspace route*: use the predicted eigenbasis to form signal/noise subspaces for MUSIC/Root-MUSIC/ESPRIT.

Angle extraction remains purely classical; the learned front-end only repairs second-order structure.

Baselines and configurations. Baselines and configurations.

We evaluate each estimator *as-is* on (i) the raw SCM $\hat{\mathbf{R}}_x$ and (ii) the reconstructed covariance $\hat{\mathbf{R}}_{\text{clean}}$:

- Bartlett and MVDR (Capon),
- MUSIC with known K ; Root-MUSIC (ULA),
- ESPRIT and Unitary-ESPRIT.

We focus our comparison on these *classical estimators* to demonstrate the plug-and-play enhancement provided by our front-end. Direct comparison with recent state-of-the-art deep learning DoA methods (e.g., [24]–[28]) is intentionally excluded, as such models typically operate within a narrow generalization regime and often require retraining or fine-tuning for each scenario. In contrast, our proposed ReconUNet employs a single frozen set of weights to handle all variations simultaneously—across SNR levels, multipath configurations, and array imperfections—without any re-optimization.

Spectrum scans use a 1° grid on $[30^\circ, 150^\circ]$ with three-point parabolic refinement around peaks. In all cases K is assumed known and identical across conditions.

Evaluation protocol and metrics. Accuracy is reported as RMSE of angular error (degrees) after optimal assignment between the K estimates and ground truth. Plots typically use a logarithmic y -axis; where informative, we overlay the Cramér–Rao bound (ideal matched model).

Scenario matrix (stress tests). Each scenario is evaluated under *perfect* and *imperfect* arrays while sweeping SNR $\in [-20, 20]$ dB:

- Basic: one direct source (no multipath).
- Moderate: two direct sources plus one coherent replica (of one source).
- Advanced 1: one direct source with six coherent replicas (OOD relative to training with $M \leq 3$).
- Advanced 2: four direct sources with three coherent replicas spread across emitters.

The 10° minimum-separation rule applies only to direct sources; coherent replicas are unconstrained.

Evaluation set construction (consistent angles across SNR).

To make SNR sweeps directly comparable, we fix the geometry across SNR levels and only change noise power. The evaluation set comprises 9,000 total samples arranged as: **9 SNR levels** ($-20, -15, -10, -5, 0, 5, 10, 15, 20$ dB) and **1,000 samples per SNR level** (the same angles at every SNR). Creation:

- 1) Draw 1,000 unique angle configurations (numbers of sources, their DoAs, and multipath layout) according to the scenario definition; freeze these indices.
- 2) For each SNR, render one sample per configuration by adjusting only σ_n^2 ; hence, *the same angles appear at every SNR*.

V. RESULTS AND DISCUSSION

Scope. We evaluate the estimators and scenarios defined in Section IV and report accuracy as RMSE of angular error (degrees). We refrain from restating the setup and refer the reader to Section IV for geometry, data generation, feature construction, baselines, and metrics.

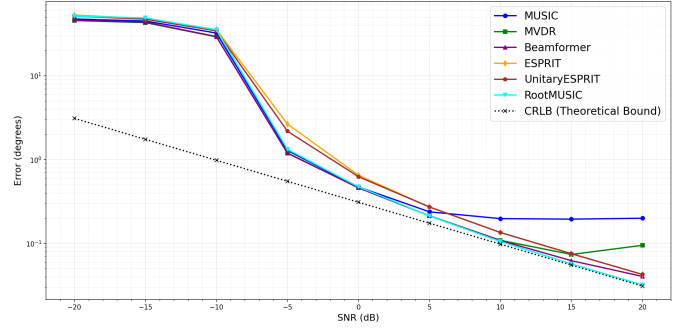


Fig. 6: Ideal-case benchmark (Basic scenario, no multipath, no imperfections): RMSE vs. SNR for classical estimators with CRLB overlay.

TABLE I: Ideal-case RMSE (deg) at Low (-10 dB), Mid (5 dB), and High (20 dB) SNR. Best per column in **bold**.

Method	Low SNR	Mid SNR	High SNR
Bartlett	27.3631	0.2071	0.0379
MVDR	26.8068	0.2080	0.0915
MUSIC	33.6959	0.2360	0.1971
Root-MUSIC	37.2366	0.2070	0.0316
(Unitary-)ESPRIT	35.7450	0.2758	0.0408

A. Ideal-Case Baseline: Classical Methods Under Matched Models

Objective. Before introducing our front-end, we establish the performance of classical estimators when the data exactly satisfy their assumptions (Basic scenario in Section IV: narrowband, no multipath, no array imperfections, known K).

Findings. In Fig. 6 we report *only* the classical estimators using the same 1° scan grid adopted throughout the paper for consistency. Under matched conditions, Root-MUSIC and (Unitary-)ESPRIT track the CRLB closely. By contrast, the *grid-scanning* methods (MUSIC, MVDR) exhibit a characteristic high-SNR flattening: as SNR increases, their spectral peaks sharpen and the combination of a discrete angle grid with three-point parabolic refinement imposes a finite angular resolution, yielding an apparent performance floor. MVDR experiences the same effect because its mainlobe narrows with SNR, amplifying sub-grid localization limits; Bartlett appears less affected since its broader peaks make the parabolic fit less sensitive to discretization. The *gridless* methods (Root-MUSIC, ESPRIT/Unitary-ESPRIT) avoid discretization entirely and therefore do not exhibit this high-SNR floor.

For completeness, we also evaluated MUSIC/MVDR with a finer 0.1° grid. As expected, this reduces discretization error. Root-MUSIC/ESPRIT curves are essentially unchanged under finer grids, as they do not rely on scanning. The finer grid increases scan cost by roughly $\times 10$ for a fixed field of view, which is a practical trade-off when targeting sub-tenth-degree accuracy in high-SNR, well-modeled conditions. Representative 0.1° results are shown in Fig. 7, and pivot statistics appear in Table I for the 1° setting.

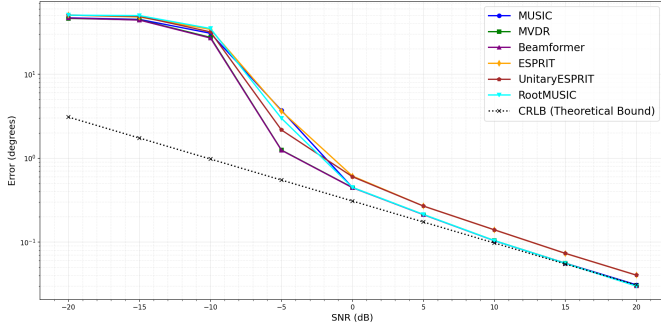


Fig. 7: Ideal-case benchmark with a finer 0.1° scan grid: RMSE vs. SNR, with CRLB overlay.

Interpretation. This baseline confirms that when assumptions match reality, classical subspace methods are already near-optimal; our learned front-end is aimed at restoring *similar* behavior in the stressed regimes of Section IV.

B. RMSE vs. SNR Across Scenarios

Objective. Using the scenarios defined in Section IV, we compare each classical estimator fed with (i) the raw SCM and (ii) our reconstructed covariance / predicted EVD, across an SNR sweep. To demonstrate *plug-and-play* compatibility, we deliberately vary the classical *subspace* back end across scenarios (MUSIC, ESPRIT, and Unitary-ESPRIT). The last panel stresses *out-of-distribution* (OOD) generalization with a single direct source and *six* coherent replicas—more multipath than seen during training.

Evaluation set construction (consistent angles across SNR). To make the SNR sweeps directly comparable, we fixed the scene geometry across SNR levels and only changed the noise power. Concretely, the evaluation set comprises 9,000 total samples arranged as:

- **9 SNR levels:** $-20, -15, -10, -5, 0, 5, 10, 15, 20$ dB;
- **1,000 samples per SNR level** (same angles at every SNR).

Evaluation dataset creation process.

- 1) **I):** draw 1,000 unique angle configurations (numbers of sources, their DoAs, and any multipath layout) according to the scenario definition. These indices are fixed and reused for all SNRs.
- 2) **II (per SNR):** for each of the same 1,000 angle configurations, render one sample at each of the 9 SNR levels by adjusting only σ_n^2 . Thus, *the same angles appear at every SNR*.

Findings. Observations from Figs. 8–10:

- **Consistency across subspace methods.** In each panel we intentionally pair our front end with a *different* classical subspace algorithm (MUSIC in Fig. 10, ESPRIT in Fig. 8, Unitary-ESPRIT in Fig. 9). The dashed curves (UNet_MUSIC, UNet_ESPRIT, UNet_UNITARY_ESPRIT) improve upon their solid counterparts over wide SNR ranges, confirming that the proposed statistics repair is *algorithm-agnostic* and integrates seamlessly with distinct subspace engines.

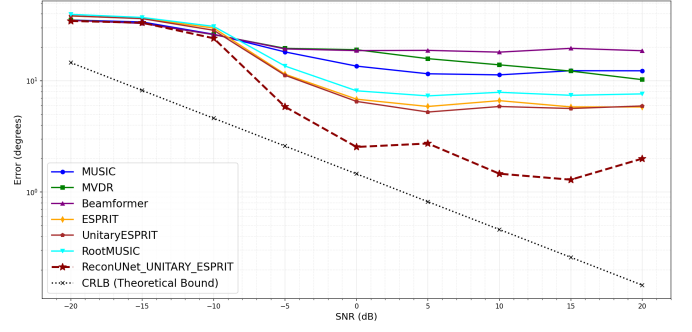


Fig. 8: RMSE vs. SNR (Moderate): 3 direct sources + 1 coherent replica; ESPRIT back end. Thin solid lines: classical on raw SCM. Dashed star-marked line: same back end on reconstructed covariance / predicted EVD.

- **Low-SNR and coherence regimes.** The largest gains appear where the signal–noise eigengap is smallest (low SNR and coherent mixtures).
- **Crowded scenes.** In the 4+3 case (Fig. 9) the dashed Unitary-ESPRIT curve remains well below its solid baseline across the sweep, indicating that the front end suppresses spurious peaks and improves assignment stability even under heavy overlap.
- **Out-of-distribution behavior at high SNR.** In the OOD setting with 1 direct source and 6 coherent replicas (Fig. 10), our method markedly improves RMSE in the low-to-mid SNR range where it was trained to operate; however, for SNRs *above* roughly 10 dB, the dashed UNet_MUSIC curve turns upward while the best classical baselines continue to decrease. This suggests a *regularization-induced floor*: the network, optimized primarily for low-SNR robustness and trained within a narrower SNR window, may over-smooth or bias the statistics in clean regimes it did not see during training. Practically, this indicates that the front end is best used where covariance estimates are noisy or ill-conditioned; at very high SNR in strongly off-distribution scenes, direct classical processing may be preferable unless the training schedule and losses are adapted for that regime.

Numerical summaries at 0 dB SNR are reported in Table II. As evident in Table II, the ESPRIT-based variant achieves the lowest RMSE across scenarios. This is expected: ESPRIT operates on the *signal* subspace—the very object our network is trained to reconstruct and align—whereas MUSIC relies on the *noise* subspace and a discrete grid, making it more sensitive to residual leakage and discretization error. In the crowded case, ESPRIT also outperforms Unitary-ESPRIT because the latter depends on forward–backward symmetry (centro-Hermiticity); array imperfections and coupling violate this structural assumption, degrading Unitary-ESPRIT more than standard ESPRIT.

C. Angle-Sweep Performance (3 Sources, No Multipath)

Setup. We sweep three ground-truth angles across the field of view at fixed separation $\Delta\theta = 10^\circ$ and 0 dB SNR, with no multipath and an imperfect array (see Section IV). For each

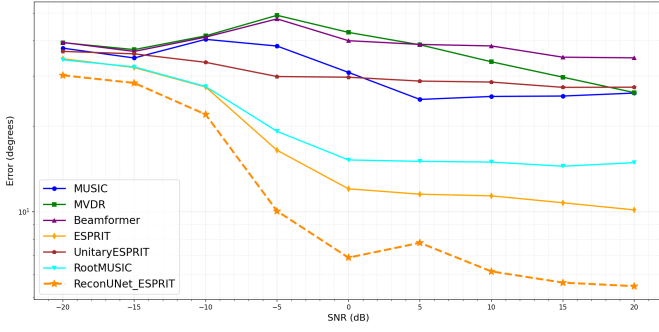


Fig. 9: RMSE vs. SNR (Crowded): 4 direct sources + 3 coherent replicas; Unitary-ESPRIT back end. Thin solid lines: classical on raw SCM. Dashed star-marked line: same back end on reconstructed covariance / predicted EVD.

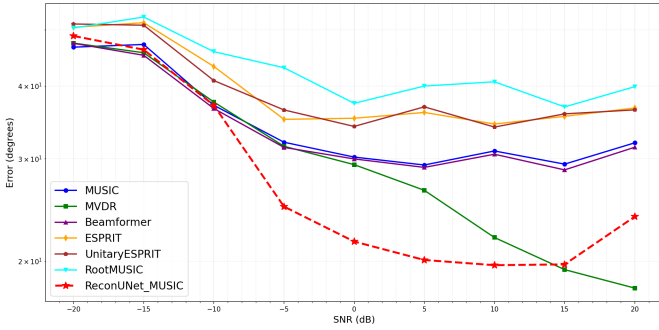


Fig. 10: RMSE vs. SNR (OOD): 1 direct source + 6 coherent replicas; MUSIC back end. Thin solid lines: classical on raw SCM. Dashed star-marked line: same back end on reconstructed covariance / predicted EVD.

TABLE II: RMSE at 0 dB SNR (deg) across scenarios. Columns: OOD (1+6), Moderate (3+1), Crowded (4+3). Best per column in **bold**.

Method	OOD (1+6)	Mod (3+1)	Crowd (4+3)
Bartlett	29.98	18.68	40.00
MVDR	29.31	19.02	42.78
MUSIC	30.20	13.55	30.93
Root-MUSIC	37.33	8.18	15.21
ESPRIT	35.22	6.83	12.03
Unitary-ESPRIT	34.11	6.52	29.75
ReconUNet + MUSIC	21.62	4.69	14.16
ReconUNet + Unitary-ESPRIT	23.29	2.54	28.29
ReconUNet + ESPRIT	20.89	2.08	6.89

method we overlay ground truth (solid lines) and the tracked estimates (markers) versus sample index. We report classical **MUSIC**, **Root-MUSIC**, and **ESPRIT**, and the same back ends when fed by our reconstructed covariance / predicted EVD.

Method-wise observations at 0 dB, $K=3$, $\Delta\theta = 10^\circ$. Beyond the aggregate trend, the six panels in Fig. 11 reveal characteristic behaviors for each classical back end and its aided counterpart:

MUSIC vs. ReconUNet + MUSIC.: The aided curve yields only *modest* improvements in trajectory smoothness and bias; in several ranges the scatter and small under/overshoots persist. Two factors explain this: (i) MUSIC estimates angles through *orthogonality to the noise subspace*, whereas our training

explicitly aligns the *signal* subspace and reconstructs a clean covariance—so the noise basis is only an implicit by-product. Small residual errors in the noise subspace can therefore survive and limit gains. (ii) MUSIC still performs a grid scan with parabolic refinement; even with a better covariance, the sub-grid interpolation error remains a bottleneck. Taken together, these effects make the aided MUSIC less improved than the gridless methods below. This also suggests a path to stronger gains: augment training with a noise-subspace loss or a MUSIC-spectrum consistency term.

Root-MUSIC vs. ReconUNet + Root-MUSIC.: The aided variant tracks the ground-truth lines tightly with visibly reduced variance and far fewer outliers in the sweep (see the zoomed insets). Root-MUSIC depends on stable polynomial roots near the unit circle; when the eigen-gap is small, noise-signal leakage perturbs the root locations and the “root selection” step can fail. The reconstructed covariance enlarges the eigen-gap and reduces leakage, so the correct roots remain dominant and smoothly evolve across angle—eliminating the sporadic slips seen with the classical SCM.

ESPRIT vs. ReconUNet + ESPRIT.: Aided ESPRIT exhibits the largest qualitative improvement: trajectories become nearly collinear with the ground-truth lines and mode-swapping events disappear in the sweep. ESPRIT estimates inter-subarray phase shifts from the *signal* subspace via a rotational-invariance eigenproblem; it is therefore directly helped by our signal-subspace alignment objective and by the Hermitian/diagonal-loading stabilization. The cleaner subspace yields more accurate eigenvalues (phase factors), restoring linear phase progression and producing smooth, bias-free tracks in the inset.

Angles near endfire.: Across all three panels of classical methods, estimates are noticeably weaker as trajectories approach the array’s endfire region ($\theta \approx 30^\circ$ or 150°): the scatter grows, small biases appear, and classical methods occasionally slip. This follows from the ULA manifold $\mathbf{a}(\theta) = [e^{-j\pi(n-1)\cos\theta}]_n$: its angular sensitivity scales with $|\partial\mathbf{a}(\theta)/\partial\theta| \propto |\sin\theta|$. Near endfire, $\sin\theta \rightarrow 0$, so the Fisher information drops, beampatterns broaden, and the signal/noise eigengap shrinks. With our repaired covariance, the endfire penalty is *consistently reduced*: the enlarged eigengap and lower subspace leakage sharpen spectra and stabilize the rotational-invariance/eigen-root steps. In Root-MUSIC and ESPRIT this mitigation is often striking—endfire degradation is faint or, within this sweep, not apparent—whereas MUSIC still shows residual endfire sensitivity due to its reliance on the **noise** subspace and grid refinement. While the front-end improves robustness, the fundamental manifold ill-conditioning near endfire remains a limiting factor.

Summary. Gridless, signal-subspace-based estimators (Root-MUSIC, ESPRIT) benefit the most from our statistics repair because the network is trained to (i) reconstruct a clean covariance and (ii) *explicitly* align the **signal** subspace. MUSIC, whose decision rule relies on the **noise** subspace and on a discrete grid, shows smaller gains—consistent with the network’s training objectives and the remaining grid interpolation limits.

Quantitatively, these trends are reflected in the per-source RMSEs reported in Table III.

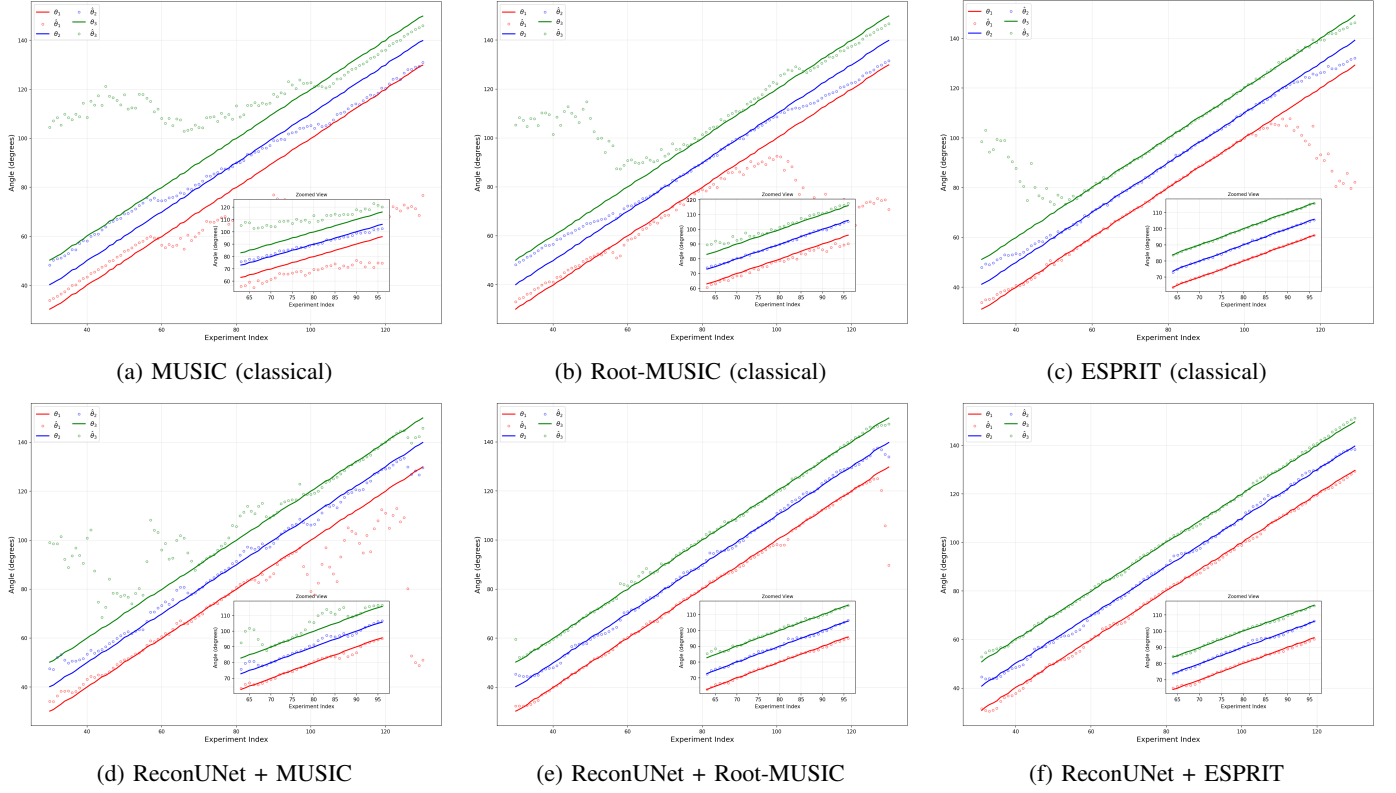


Fig. 11: Angle-sweep at 0 dB SNR with $K=3$ and $\Delta\theta = 10^\circ$ (no multipath). Top: classical back ends. Bottom: same back ends using the reconstructed covariance / predicted EVD

TABLE III: Per-source RMSE at 0 dB SNR (degrees) for $K=3$ and $\Delta\theta = 10^\circ$. Best per column in **bold**.

Method	θ_1	θ_2	θ_3
MUSIC	28.01	6.74	28.98
ReconUNet + MUSIC	17.98	3.65	16.43
Root-MUSIC	23.15	4.86	23.27
ReconUNet + Root-MUSIC	4.74	1.41	1.37
ESPRIT	13.40	2.75	13.90
ReconUNet + ESPRIT	1.08	1.18	0.89

D. Spectrum Comparison at 0 dB SNR

Setup. We visualize (top row) the **MUSIC** pseudospectrum and (bottom row) the **Root-MUSIC** polynomial roots on the unit circle at 0 dB, for two direct sources at fixed angles $\Theta = \{90.6^\circ, 80.1^\circ\}$. Columns correspond to coherent replica counts $\text{mp} \in \{0, 1, 2\}$. The multipath departure angles in each panel are listed as Θ_{mp} .

Findings.

- **MUSIC.** Relative to spectra computed from the raw SCM, the spectrum obtained with our reconstructed covariance exhibits *higher* and *narrower* mainlobes at the true angles, with *deeper* nulls against the noise subspace. Formally, this reflects a larger signal–noise eigengap, which reduces $\|\mathbf{E}_n^H \mathbf{a}(\theta)\|_2$ near the true θ and thus raises $P_{\text{MUSIC}}(\theta) = 1/\|\mathbf{E}_n^H \mathbf{a}(\theta)\|_2^2$. Coherent replicas ($\text{mp}=1,2$) that introduce shoulders or spurious peaks in the raw

spectrum are *suppressed* after repair, indicating reduced subspace leakage from correlated paths and improved orthogonality between $\mathbf{a}(\theta)$ and the reconstructed noise basis.

- **Root-MUSIC.** With the repaired covariance, the two DoA-corresponding roots remain *closer to the unit circle* and *stable* across scenarios, while replica-induced roots cluster more interior. This is consistent with the perturbation view: enlarging the eigengap decreases the sensitivity of the signal roots to noise/coherence and stabilizes the root-selection step. In the raw case, coherent replicas can pull the physical roots inward or create ambiguous clusters; after repair, the physical roots dominate and lie closer to the unit circle at arguments matching Θ .

E. Synthesis and Mechanism: Why the Gains Appear

Summary. Across Section V-A–Section V-D, a consistent picture emerges: replacing the raw SCM with our reconstructed covariance (or using the predicted EVD) *increases the effective SNR and the signal–noise eigengap*, which in turn sharpens spectra, stabilizes gridless solvers, and reduces RMSE. This behavior is robust across back ends (MUSIC/Root-MUSIC/ESPRIT), scene complexity (from basic to crowded), and even in out-of-distribution (OOD) multipath—with the noted caveat at very high SNR in the OOD case.

Model view. Let the measured covariance be

$$\hat{\mathbf{R}}_x = \underbrace{\mathbf{A}\mathbf{R}_s\mathbf{A}^H}_{\text{ideal target } \mathbf{R}^*} + \sigma_n^2\mathbf{I} + \underbrace{\Delta_{\text{mp}} + \Delta_{\text{imperf}}}_{\text{coherence, mismatches}}$$

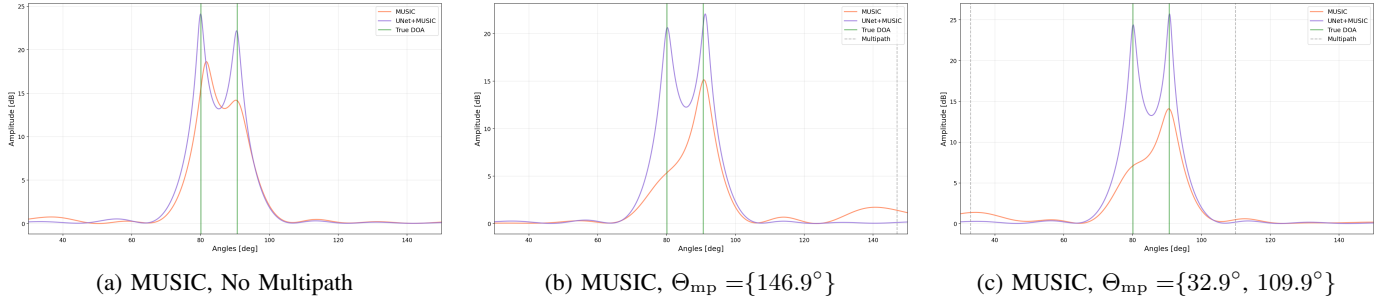


Fig. 12: Spectral views at 0 dB SNR for two sources at $\Theta = \{90.6^\circ, 80.1^\circ\}$ with coherent replicas $mp \in \{0, 1, 2\}$. MUSIC pseudospectra.

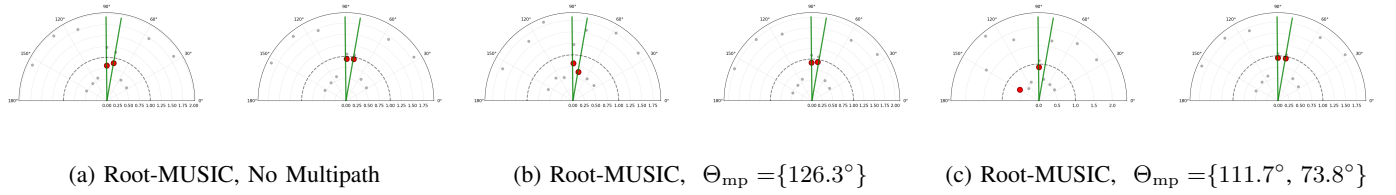


Fig. 13: Root constellations on the complex plane at 0 dB SNR for two sources at $\Theta = \{90.6^\circ, 80.1^\circ\}$ with coherent replicas $mp \in \{0, 1, 2\}$. Root-MUSIC polynomial roots on the unit circle.

where Δ_{mp} captures coherent multipath (rank collapse and cross-terms), Δ_{imperf} encodes array mismatches (gain/phase offsets, position jitter, mutual coupling). The network learns a mapping from the lag-stack features to a *clean, Hermitian, well-conditioned* surrogate

$$\hat{\mathbf{R}}_{EVD} \approx \mathbf{R}^*,$$

and/or directly to its eigendecomposition. In effect, the corruptions are suppressed (denoised/de-biased), so the top- K signal eigenvalues separate more clearly from the $(N - K)$ noise eigenvalues. A convenient proxy for this improvement is the *effective SNR*

$$\text{SNR}_{\text{eff}} \triangleq \frac{\sum_{i=1}^K \lambda_i(\hat{\mathbf{R}})}{\frac{1}{N-K} \sum_{i=K+1}^N \lambda_i(\hat{\mathbf{R}})},$$

which increases when Δ_{mp} , Δ_{imperf} are attenuated.

One model for many scenes. Because the network learns to *repair statistics*—rather than regress angles or overfit a particular back end—it generalizes across the scenario matrix with a single set of weights. No per-scenario fine-tuning is required, and the same front end plugs into distinct subspace engines (MUSIC, Root-MUSIC, ESPRIT), as evidenced by Section V-B.

Practical takeaway. Learning to denoise and de-bias the covariance/subspace restores the conditions under which classical DoA estimators excel. The net effect is a higher SNR_{eff} and larger eigengap, which translate into lower RMSE, fewer slips, and improved stability across methods and scenes—while preserving interpretability and the drop-in character required for real systems.

VI. CONCLUSION

This work introduced a subspace-guided learning front end that reconstructs a clean covariance (and its eigendecomposition) from corrupted, low-SNR snapshots, and then hands those statistics to off-the-shelf DoA estimators without changing their internals. Across the operating space in Section IV—including coherent multipath and realistic array imperfections—the same trained model consistently tightened error curves and reduced slips for MUSIC, Root-MUSIC, ESPRIT, and Unitary-ESPRIT. The gains are most pronounced where the signal/noise eigengap is small (low SNR, coherence, imperfect arrays), and they persist under crowded scenes; by design, the approach remains *algorithm-agnostic* and *plug-and-play*.

A central outcome is *generalization across scenarios with a single set of weights*. We deliberately varied the downstream subspace algorithm per scenario (Section V-B) and observed similar benefits, demonstrating that the front end learns to repair second-order structure rather than overfitting to a particular spectrum or solver. In other words, we do not require per-scenario or per-algorithm fine-tuning: the same model operates from the basic single-source regime through multi-source coherent mixtures with array mismatch—even when the exact replica counts or angular configurations differ from training.

The results further indicate practical *field suitability*. Our supervision targets are theoretical, closed-form objects (the clean covariance and its EVD on the nominal manifold). These targets exist for any real scene once a nominal geometry is specified; hence, in real data pipelines they can be derived from physics rather than annotation, enabling semi-synthetic or weakly supervised training: render corrupted inputs from measured snapshots, supervise toward the nominal clean statistics, and deploy the repaired outputs within standard subspace engines. This preserves interpretability (explicit EVD)

and enables classical confidence diagnostics (e.g., eigengap-based quality checks) without requiring end-to-end angle labels.

Practical considerations.: At inference, the method adds one forward pass over a compact lag-sensor tensor. Users can either (i) feed the reconstructed covariance to any spectrum-scanning back end or (ii) bypass runtime EVD entirely by using the predicted eigenbasis. Because outputs are classical objects (SCM/EVD), integration with existing peak pickers, source-enumeration heuristics, and downstream tracking is straightforward.

Limitations and lessons.: Two phenomena bound performance. First, near endfire the ULA manifold is intrinsically ill-conditioned; while the front end mitigates variance by enlarging eigengaps, the geometry still limits ultimate accuracy. Second, in the out-of-distribution high-SNR corner (Section V-B, 1+6 replicas), we observed a shallow floor for the aided MUSIC. These suggest (i) explicit endfire-aware augmentation and (ii) training objectives that also stabilize the *noise* subspace or impose spectrum-consistency penalties for grid-based methods.

Outlook.: Further work will extend training to a broader SNR range and to richer combinations of source counts and coherent-multipath replicas, so the model encounters a denser, more challenging operating envelope. We will also investigate generalization across array geometries (e.g., ULA, UCA, and nonuniform apertures). Finally, we will validate the approach on real measurement data, comparing against calibration-grade references to characterize failure modes, tune losses, and assess how well the learned statistics transfer from synthetic supervision to field data.

Takeaway. Learning to *repair the statistics*—instead of regressing angles end-to-end—yields a single, reusable model that generalizes across scenarios and slots seamlessly into classical DoA pipelines. The approach improves accuracy where it matters operationally, preserves estimator transparency, and offers a clear path from theoretical targets to real-world use.

REFERENCES

- [1] H. Krim and M. Viberg, “Two decades of array signal processing research: The parametric approach,” *IEEE Signal Processing Magazine*, vol. 13, no. 4, pp. 67–94, 1996.
- [2] P. Stoica and R. L. Moses, *Spectral Analysis of Signals*. Upper Saddle River, NJ: Prentice Hall, 2005.
- [3] P. Stoica and A. Nehorai, “Music, maximum likelihood, and cramér-rao bound,” *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 37, no. 5, pp. 720–741, 1989.
- [4] B. D. Van Veen and K. M. Buckley, “Beamforming: A versatile approach to spatial filtering,” *IEEE ASSP Magazine*, vol. 5, no. 2, pp. 4–24, Apr. 1988.
- [5] J. Capon, “High-resolution frequency-wavenumber spectrum analysis,” *Proceedings of the IEEE*, vol. 57, no. 8, pp. 1408–1418, Aug. 1969.
- [6] R. O. Schmidt, “Multiple emitter location and signal parameter estimation,” *IEEE Transactions on Antennas and Propagation*, vol. 34, no. 3, pp. 276–280, Mar. 1986.
- [7] R. Roy and T. Kailath, “Esprit—estimation of signal parameters via rotational invariance techniques,” *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 37, no. 7, pp. 984–995, Jul. 1989.
- [8] M. Haardt and J. A. Nossék, “Unitary ESPRIT: How to obtain increased estimation accuracy with a reduced computational burden,” *IEEE Transactions on Signal Processing*, vol. 43, no. 5, pp. 1232–1245, May 1995.
- [9] M. Haardt and M. E. Ali-Hackl, “Unitary esprit: How to exploit additional information inherent in the rotational invariance structure,” in *Proc. IEEE Int. Conf. on Acoustics, Speech, and Signal Processing (ICASSP)*, 1994, pp. IV-229–IV-232.
- [10] A. J. Barabell, “Improving the resolution performance of eigenstructure-based direction-finding algorithms,” in *Proc. IEEE Int. Conf. on Acoustics, Speech, and Signal Processing (ICASSP)*, Boston, MA, 1983, pp. 336–339.
- [11] H. Wang, K. J. R. Liu, and H. Anderson, “Spatial smoothing for arrays with arbitrary geometry,” in *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Adelaide, Australia: IEEE, 1994, pp. IV-509–IV-512.
- [12] C. Qi, Y. Wang, Y. Zhang, and Y. Han, “Spatial difference smoothing for doa estimation of coherent signals,” *IEEE Signal Processing Letters*, vol. 12, no. 11, pp. 800–803, Nov. 2005.
- [13] S. U. Pillai and B. H. Kwon, “Forward/backward spatial smoothing techniques for coherent signal identification,” *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 37, no. 1, pp. 8–15, Jan. 1989.
- [14] B. Friedlander and A. J. Weiss, “Direction finding in the presence of mutual coupling,” *IEEE Transactions on Antennas and Propagation*, vol. 39, no. 3, pp. 273–284, mar 1991.
- [15] A. J. Weiss and B. Friedlander, “On the cramér-rao bound for direction finding of correlated signals,” in *Proc. Asilomar Conf. on Signals, Systems and Computers (ACSSC)*, Pacific Grove, CA, Nov. 1990, pp. 941–945.
- [16] Z. Zhang, Z. Shi, and Y. Gu, “Ziv-zakai bound for doa estimation,” *arXiv preprint arXiv:2209.03700*, 2022.
- [17] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-based learning applied to document recognition,” *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [18] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [19] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, “Model-based deep learning,” *arXiv preprint arXiv:2012.08405*, 2022, v3.
- [20] J. Wolff, S. Schultz, P. A. Hoeher, C. Waldschmidt, D. Wübben, and A. Dekorsy, “Deep learning-based beamforming: A review of the state of the art and future challenges,” *IEEE Access*, vol. 10, 2022.
- [21] Z.-M. Liu, C. Zhang, and P. S. Yu, “Direction-of-arrival estimation based on deep neural networks with robustness to array imperfections,” *IEEE Transactions on Antennas and Propagation*, vol. 66, no. 12, Dec. 2018.
- [22] L. Wu, Z.-M. Liu, and Z.-T. Huang, “Deep convolution network for direction of arrival estimation with sparse prior,” *IEEE Signal Processing Letters*, vol. 26, no. 11, pp. 1688–1692, Nov. 2019.
- [23] A. M. Elbir, “Deepmusic: Multiple signal classification via deep learning,” *IEEE Sensors Letters*, vol. 4, no. 4, p. 7001004, Apr. 2020.
- [24] P. Chen, Z. Chen, L. Liu, Y. Chen, and X. Wang, “Sdoa-net: An efficient deep learning-based doa estimation network for imperfect array,” *IEEE Transactions on Instrumentation and Measurement*, 2024, early access.
- [25] D. H. Shmuel, J. P. Merkofer, G. Revach, R. J. G. van Sloun, and N. Shlezinger, “Subspacenet: Deep learning-aided subspace methods for doa estimation,” *arXiv preprint arXiv:2306.02271*, Jul. 2024.
- [26] J. P. Merkofer, G. Revach, N. Shlezinger, T. Routtenberg, and R. J. G. van Sloun, “DA-MUSIC: Data-driven DoA estimation via deep augmented MUSIC algorithm,” *IEEE Transactions on Vehicular Technology*, vol. 73, no. 2, pp. 2771–2785, Feb. 2024.
- [27] G. K. Papageorgiou, M. Sellathurai, and Y. C. Eldar, “Deep networks for direction-of-arrival estimation in low snr,” *IEEE Transactions on Signal Processing*, vol. 69, 2021.
- [28] B. Zhou, K. Xu, Y. Quan, and M. Xing, “A deep learning-based supervised transfer learning framework for doa estimation with array imperfections,” *arXiv preprint arXiv:2504.13394*, Apr. 2025.
- [29] A. M. Alam, C. O. Ayna, S. Biswas, J. T. Rogers, J. E. Ball, and A. C. Gurbuz, “Deep learning-based direction-of-arrival estimation with covariance reconstruction,” in *2024 IEEE Radar Conference (RadarConf24)*, 2024.
- [30] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, ser. Lecture Notes in Computer Science, vol. 9351. Springer, 2015, pp. 234–241.